

High-Reach Mechanical Scraper: Engineering an Accessible Solution for Spotted Lanternfly Egg Mass Removal

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Context and Problem Statement

Spotted Lanternfly (SLF) egg masses represent a critical vector for the invasive species' proliferation, posing a severe economic threat to viticulture and agricultural industries. Vineyard managers attempt to eradicate these populations during the dormant season to disrupt the lifecycle before spring hatching. In fact, many agricultural organizations, such as the U.S. Department of Agriculture, advise people to remove their egg masses. However, approximately 90% of SLF egg masses are deposited 8 to 10 feet above the ground, rendering them completely inaccessible without a ladder. This makes manual scraping hazardous, physically taxing, and highly inefficient for agricultural workers who must navigate uneven terrain.

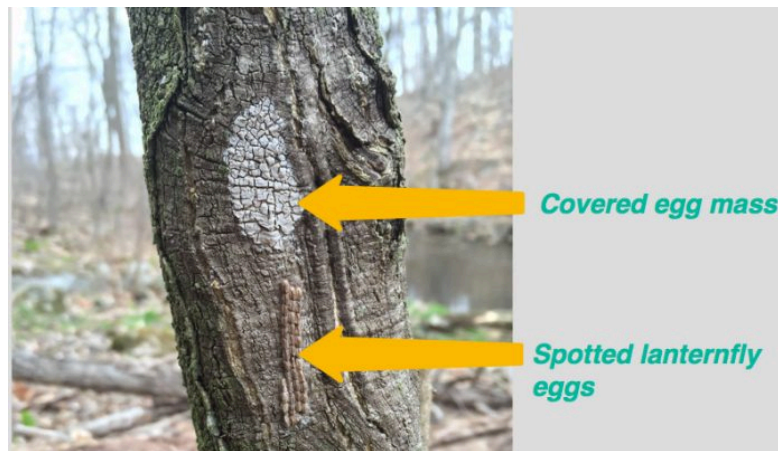


FIGURE 1: Photo of high-reach SLF egg masses on bark, showing the camouflage and height scale

Our team focused strictly on engineering a ground-based mechanical removal tool to safely target this high-reach demographic. The primary constraints shaping our design were user safety through the elimination of ladder use, ergonomics to manage the heavy moment arm generated by a long pole, and efficacy to achieve high removal rates without causing host-plant damage.

Final Prototype and Application

We engineered a high-reach, lightweight, pole-mounted mechanical scraper designed to safely extend a user's operational reach by over 6 feet. The system architecture features three main subsystems: a locking telescoping aluminum pole, a custom 3D-printed scraper head with an integrated sheet metal scraping edge, and an ergonomic handle and brace system.

To safely manage the high rotational torque generated by scraping at the end of a 10-foot lever arm, the base incorporates a rigid T-connector forearm brace lined with weather-resistant closed-cell foam and secured with a quick-release buckle strap. In practice, a worker visually locates an out-of-reach mass, adjusts the telescoping pole to the appropriate height, and applies an upward scraping motion. The rigid sheet metal edge cleanly detaches the eggs from the bark, and an integrated lower bowl geometry reliably captures the dislodged debris, entirely bypassing the need for dangerous ladders or cumbersome ground catch-tarps.

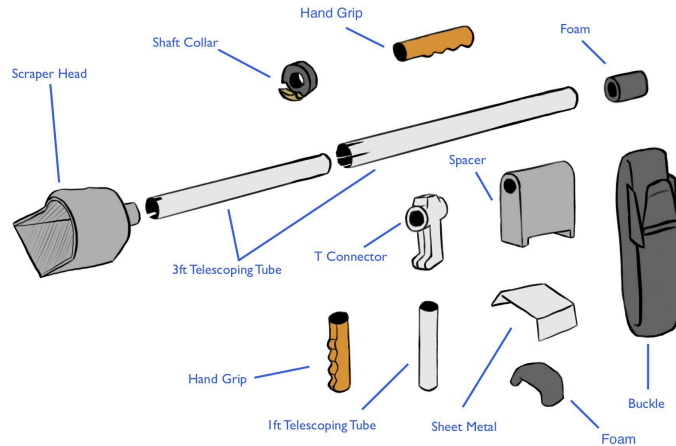


FIGURE 2: Exploded Assembly showing the components of the final prototype



FIGURE 3: Assembled Final Prototype

Testing and Results

We conducted multiple rounds of prototyping and testing to reach our final design and satisfy our success criteria. We started with our first prototype, evaluating bracing methods and the ideal weight balance for a user. We found that a cylindrical forearm brace, providing support both underneath and over the arm, significantly reduced the moment required for a user to apply and made a six-foot pole easy to handle within a reasonable weight parameter (less than 10 pounds). Our second prototype mainly tested the mechanics of the scraper. A 3D-printed custom part was designed to both scrape and store large quantities of egg masses, shaped almost like a mailbox. We performed our testing by placing 10 egg masses (we modeled with clay) at various heights on the wall and a couple of egg masses on sandpaper; because the clay analog is heavier, denser, and stickier than actual SLF egg masses, real-world performance on dry egg masses is expected to be even higher. The results of our testing helped us determine that the scraper was at least 90% effective by mass in removing the clay, but the leading edge of the 3D-printed plastic scraper quickly eroded and needed to be improved. Our final prototype design featured the telescoping mechanism and a new (interchangeable) scraping edge made of aluminum sheet metal. The prototype passed with a final weight of exactly 2.85 pounds, implying a highly manageable moment arm that should minimize shoulder fatigue for a user. In terms of edge durability, we found less than 5 millimeters of material loss after 40 scraping cycles of testing (repeated from our second prototype—egg masses placed on the classroom walls and sandpaper). And, the new scraper edge was just as effective in removing material, and the prototype as a whole demonstrated no evidence of structural deformation at various lengths, implying the success of our telescoping mechanism.



FIGURE 4: First Functional vs Final Prototype Scraper Efficacy and Durability Comparison
Prototype, Fabrication, and Bill of Materials

The final prototype was fabricated using a combination of custom and off-the-shelf components. The scraper head and brace spacer were modeled in CAD and 3D-printed in PLA, while all other components were found on McMaster-Carr. In the final iteration, the scraper's leading edge was replaced by a manually folded aluminum sheet metal insert, directly addressing the edge erosion observed in earlier plastic prototypes. This sheet metal was press-fit into a slot in the head, and the head was press-fit to the smaller diameter telescoping rod. This rod was secured to the other larger rod with a quick-release shaft collar, which was connected to the handle via a tee through-hole connector. The forearm brace assembly consisted of the 3D-printed spacer, bent sheet metal, EVA foam padding, and a nylon strap with a quick-release buckle. These parts were all appended together using high-strength adhesive. The total material cost was \$263.56 of a \$350 budget, with the shaft collar and tee connector representing the largest individual expenses. In a production context, in-house fabrication of these components would substantially reduce unit costs.

Conclusion and Recommendation

Based on rigorous quantitative lab testing, the prototype successfully achieved all primary design goals and is currently deemed field-ready. The telescoping mechanism proved mechanically robust under simulated field stresses, and the integration of a sheet metal edge drastically improved both detachment efficacy and long-term durability over our initial plastic iterations. Additionally, the device was effective and easy to hold at the maximum operational reach. This shows that the model has met our success criteria regarding scraper efficacy, durability, weight, and reach.

The current prototype was developed at a total cost of \$263.56. While this figure represents the investment required for a functional, custom-fabricated unit, it is important to contextualize this cost within the standard development cycle of a mechanical assembly. We anticipate a significant reduction in the per-unit cost as the project transitions from the prototyping phase to small-scale or industrial production. Economies of scale, increased manufacturing optimization, and better material efficiency will easily cut the costs of this device to make it a viable product with a similar price to other farm tools.

With all that being said, the prototype is ready to enter field testing in more realistic conditions, and to be modified with more input from clientele who know what they want from the device. We recommend proceeding directly to live field trials in outdoor agricultural conditions to acquire real-world efficacy data against actual SLF egg masses and to monitor user fatigue over a full work shift. Future development should focus on minor geometric iterations to the scraper head to optimize it for highly irregular host-plant surfaces, as well as the development of modular attachments, such as localized chemical sprayers or wire brushes, to expand the tool's functional scope across different seasons.

References

- [1] NJ.gov, "Spotted Lanternfly Management." Available: <https://www.nj.gov/agriculture/divisions/pi/prog/pests-diseases/spotted-lanternfly/>
- [2] The Spotted Lanternfly Project, "How to Remove Spotted Lanternfly Eggs." Available: <https://lanternflies.org/removal/>
- [3] Usda.gov "Spotted Lanternfly." Available: <https://www.aphis.usda.gov/plant-pests-diseases/slf>

Bill of Materials

Description	McMaster Code	Quantity	Unit of measurement	Total Cost
(Length=3ft) 6063 Aluminum Round Tube Telescoping, 0.058" Wall Thickness, 1" OD 3ft	6556N114	1	Each	\$25.12
(Length=1ft) 6063 Aluminum Round Tube Telescoping, 0.058" Wall Thickness, 1" OD	6556N114	1	Each	\$10.05
(Length=3ft) 6063 Aluminum Round Tube Telescoping, 0.058" Wall Thickness, 7/8" OD	6556N113	1	Each	\$23.85
Finger-grooved Hand Grip (1" diameter)	5271T13	1	Pack of 2 each	\$11.88
Heavy Duty Aluminum Clamp-on Framing Fitting, Tee Through-Hole Connector for 1" Rail OD	9578T11	1	Each	\$44.34
Tear-Resistant Super-Cushioning EVA Foam Sheet, 1/2" Thick, 12" x 12"	86095K44	2	Each	\$26.82
Multipurpose 6061 Aluminum Sheet, 0.050" Thick, 6" x 24"	89015K175	2	Each	\$31.64
Quick-Release Clamping Shaft Collar for 1" Diameter	1511K22	1	Each	\$48.10
Light Duty Polypropylene Webbing, 2" Wide, Black	8852T64	10	ft	\$5.30
Light Duty Buckle for 2" Maximum Webbing Width, Cam	29705T31	1	Pack of 5	\$12.66
3D Printed Scraper Head		1		\$11.35
3d Printed Brace Spacer		1		\$12.45
Total Spent				\$263.56